

Received October 3, 2019, accepted November 3, 2019, date of publication November 26, 2019, date of current version December 31, 2019.

Digital Object Identifier 10.1109/ACCESS.2019.2956020

Detection of Schistosomiasis Factors Using Association Rule Mining

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ABSTRACT Bilharzia or schistosomiasis is one of the most fatal and factitious disease happens through pollute which become a significant reason of deaths in the world. Prediction and factors identification that become causes of disease in early stage, may escort to treatment before it becomes critical. Data mining techniques are used to assist medical professionals effectively in diseases' classification. This research investigates the recovery and death factors which contributes to schistosomiasis disease preprocessed dataset, collected from Hubei, China. A computerized learning method, association rule mining (Apriori) is used to spot factors. Different tools were used for analysis and model evaluation with minimum support and minimum confidence indicated higher than 90% to generate rules. In addition, attributes indicating recovery and death of individuals were identified. Strong associations of disease factors; BMI, viability, nourishment, extent to ascites etc. determined and classified through Apriori algorithm. Further, results generated by association rule mining method may useful for professionals in treatment decision with better precision.

INDEX TERMS Schistosomiasis, association rule mining (ARM), feature extraction, prediction, recovery, death, data mining.

I. INTRODUCTION

From several neglected tropical diseases (NTDs), schistosomiasis is one of the major fatal and infectious NTD of the world, also known as bilharzia or snail fever. which is a parasitic disease fastened with skin through contaminated fresh water, penetrate human body, transmit in it and migrates into human veins where parasites lay eggs which becomes causes of two stages (chronic and acute) of schistosomiasis [1], [2]. According to a report of world health organization (WHO) on schistosomiasis in 2016, out of 206.4 million infected people, only 89 million people were treated with praziquantel (drug treatment). Worldwide, 240 million population and approx. 700 million individuals of endemic countries are infected by schistosomiasis. Furthermore, 78 countries reported as transmission of disease [1], [3], [4]. In 2017, 743 million population of the world treated with preventive chemotherapy for disease elimination from different counties which indicated the recovery of 68% school aged (75% recovery from 36 countries), 68.8% soil transmitted and 69% preschool children (75% recovery from 29 countries) [5].

The associate editor coordinating the review of this manuscript and approving it for publication was Muhammad Asif Naeem.

Throughout decades, several fatal life-threatening diseases always get an attention for professionals to precept them through computerized methods [6]. Diagnosis of *S. urogenital* (one species) is performed through paper, nylon or filtration of polycarbonate and *S. haematobium* (another species) examined through analyzing blood in urine [7]. Other methods are also used in poor stricken areas for diagnostic purposes such as; immunological and serological tests [8], [9].

Volume of medical data is increasing continuously. Emerging fields such as; business, science and significantly in medical, make dataset complex, meaningful and expansive and expanded all over the world. Due to these huge volumes of data sets, identification of necessary hidden knowledge, prediction of diseases, statistical analysis, extraction of hidden information, identification of factors causing the disease is now possible and significant in today's era. Methods for discovering hidden information through evaluation of novel techniques are known as data mining techniques [9]–[11]. Significance of data mining methods may perceive as the prediction of disease schistosomiasis which is very helpful for decision makers as well as for researchers [12]. Different data mining techniques such as; decision tree, K-Nearest Neighbor (KNN), Naive Bayes, were used in healthcare [13], [14].

TABLE 1. Comparison of existing techniques.

Name	Year	Objective	Techniques	Results			Data Collection	Ref.
				Comparison	Accuracy	ARM		
Z. Liu <i>et al.</i> ,	2015	To improve the performance	Fuzzy Rule Based Method	NA	90%	✗	2003-2009	[15]
M. U. Angela & A. M. Oluwatosin	2016	Comparison of system generated results with Weka tool	Neural Network Algorithm for Classification	System generated	NA	✗	UCI Machine Learning Repository	[16]
H. Zhu <i>et al.</i> ,	2016	To identify which group in correlate with other	Clustering (multiple files of data)	Marshland & Lake areas	NA	✗	2009-2014	[17]
A. Deol, <i>et al.</i> ,	2016	To detect disease in school age children of different counties	Markov model (define matrix from A to D)	Observed data matched with cross sectionals data	NA	✗	2003-2006 2004-2007	[18]
S. S. Ranhotra	2017	To check severity of urine color and intestine of a person	Image Processing (textural technique)	Correlation of Urine color with other parameters	NA	✗	750 school children	[19]
S. Xia <i>et al.</i> ,	2017	to check the pattern analysis of schistosomiasis disease	K-Mean algorithm, Clustering	Members of disease declined yearly	NA	✗	2009-2013	[6]
D. Asarnow & R. Singh	2018	To detect schistosomiasis mansoni disease on image scale data	Asarnow-Singh algorithm for image and SVM for classification	NA	90.6%	✗	4256 Parasitic Patients	[20]
A. S. Ashour <i>et al.</i> ,	2018	To detect the effective predictive model among others	SVM, KNN, Decision Tree, other Classifiers	Linear Discriminant SVM	90%	✗	Images Data	[12]
G. Li, <i>et al.</i> ,	2018	to detect advanced schistosomiasis in the people of Hubei province	Artificial Neural Network, Decision Tree and Linear Regression	Artificial Neural Network	86.60%	✗	4136 Patients	[4]
B. Kassé <i>et al.</i> ,	2019	Monitoring system (IoT) for disease detection	ANN, SVM, RF	SVM	100%	✗	Wireless Based System	[21]

Detection of disease factors are very important. Without knowing the disease, proper medication is difficult task for epidemiologist. Identification of associated factors with disease are significant. In addition, surveys were conducted and image-based techniques were used for detection of schistosomiasis disease with no factors associated with it. Some factors like occupation, annual income and some minor diseases vary a lot but many others factors are still necessary to be known. Previous research also showed that, sanitation of water in rural areas, occupation, annual income and previous treatment methods are major factors of disease but some associated factors like splenectomy, ascites, extent of ascites and BMI are still unknown. Impact of geographical and socioeconomic factors are still unclear for the survival of affected patients.

Schistosomiasis disease is a root cause of many other diseases like liver problems, chronic stages of abdominal pain, kidney damage, enlargement in spleen etc. Therefore, identification of factors interconnected with disease are important to eradicate at early level. This research work indicated the evaluation of data mining model to identify the most contributing factors associated with disease for proper medication and treatment at early stage. For such work, understanding the strong functionalities of data mining techniques are mandatory.

The main objective of this research work is to deeply investigate the disease, assess associations between disease factors, and deeply analysis the disease using association rules in patients with schistosomiasis. Below Table 1 describes the comparison of some existing data

mining techniques used for screening prediction of disease schistosomiasis.

II. RELATED WORK

Liu *et al.* [15], performed prediction through evaluation of fuzzy rule-based method on data, collected through GPS tracker and gradually divided on yearly basis to detect disease (hupensis snail) prevalence. Angela and Oluwatosin [16] applied comparison of expert software programs using Neural Network as giving input $x(\text{value}) \rightarrow f(x)$ as output. System generated results observed as better prevalence of disease rather than other. Zhu *et al.* [17] created a database to save record of individuals and evaluates clustering algorithm on multiple files of data to identify correlation of groups among each other which shows appearance of schistosomiasis japonicum infection in marshland and lakes nearby river Yangtze. Deol *et al.* [18] developed Markov model to examine disease appearance through defined matrices from A to D consisting yearly data in them which showed matched results with cross-sectional study, indicated the prevalence *s. mansoni* and some correlation of data among each other. Ranhotra [19] worked on image processing and textural analysis techniques, and established correlation by identifying the appearance of disease through checking the severity of urine color and intestine of individuals on school aged children data. Result indicates as, the entropy of components decreases, urine samples gradually increases. Xia *et al.* [6] applied K-Mean algorithm (parameter from $\times 1$ to $\times 4$ and mean distance 1 to -1) on water, cattle yards, contaminated water snail area, population and village location data to analyses pattern of schistosomiasis disease which specifies the gradually decrease of schistosomiasis infections on yearly basis. Asarnow and Singh [20] applied Asarnow-Singh algorithm to identify *s. mansoni* through extraction of images' features by defining threshold values on image's infected foreground and background parasitic areas. Further, support vector machine (SVM) evaluated into training and testing record which provides effective results. Ashour *et al.* [12] specified level of schistosomiasis

images and extracted features by giving statistical names of legion area. Further, evaluates SVM, KNN, DT which shows that, linear discriminant svm classifier results are better than quadratic svm. Li *et al.* [4] classified into two groups (poor and advanced schistosomiasis). Moreover, they divided data into two groups as training and testing record and applied ANN, DT and LR to determine the best results among all. Confusion matrix indicated the best prediction of disease is performed by ANN model than others. Kassé *et al.* [21] developed an IoT based monitoring system for prediction and controlling of disease and collected data through wireless based sensors network. They applied different data mining algorithms for disease detection and transmission which shows the SVM shows detection of anomalies better than other models.

By going through literature survey on disease schistosomiasis, it was observed that, performance of classifiers was compared and different techniques were used to increase classifiers performance. Further, in comparison of existing techniques, work on the prediction of disease is done yet with collection of all attributes in a dataset which only gives classifier accuracy of overall system. Before treatment, knowing the disease and its affected features are examined by professionals which may use to eliminate disease at start before it become critical. However, in existing studies, neither features' relations nor clear classified rules are yet defined on which clinical attributes of patients of schistosomiasis disease can be predicted. Therefore, we purpose a method for identification of features' relations and efficient classification of schistosomiasis disease in order to better assist medical professionals.

III. METHODOLOGY

Methodology included the identification of factors' relationship using association rule mining (ARM). Proposed study follows the below Figure no 1. We preprocessed our collected data into categorical form for identification of rules after evaluation.

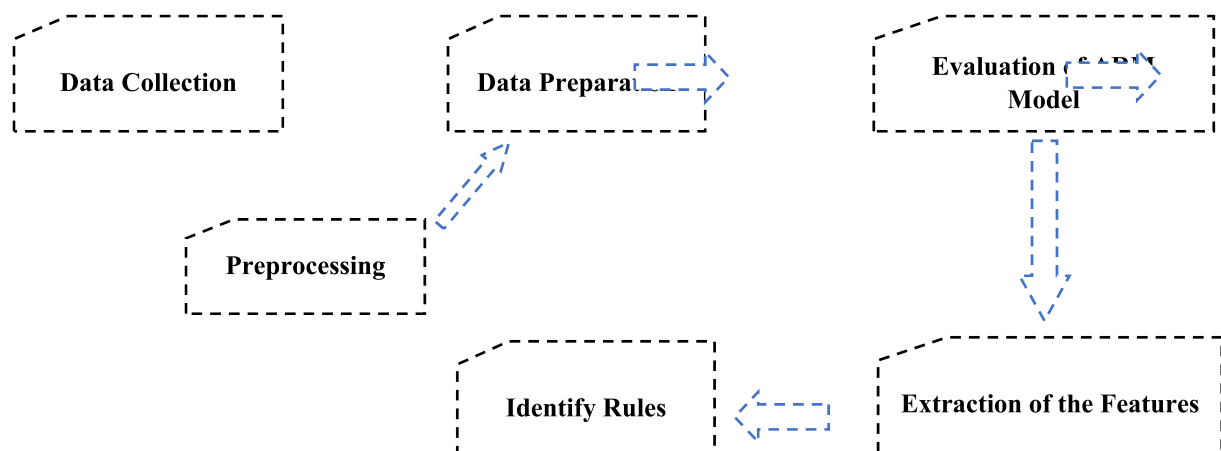


FIGURE 1. Diagram of proposed work.

A. DATA SET

Schistosomiasis data set [4] included the record of 4136 (2674 male and 1462 female) patients, diagnosis with advance schistosomiasis disease in them. Data set included $16+1(\text{Label}) = 17$ attributes. Two groups defined for class variable with binary values as 0 or 1 into recovery and death respectively. Data consist of demographic and socio-economic attributes, clinical factors, surgical treatment methods, and class label etc. Description of data is defined in and Table 2 and 3 below.

TABLE 2. Dataset description.

Number of Instances	Recovery or Improvement	Death	Total
	1232 (29.8%)	2904 (70.2%)	4136
No. of Attributes	16 + 1 (Label) = 17 Features		
Class Label (Classification)	Recovery and Death		

B. PREPROCESSING

Data mining functionalities vary on reliability and quality of data. Raw data indicates the missing values, noisy data, inconsistency and outliers in it [14]. In addition, we used preprocessing of data on our data set in multiple steps such as; first, collected data and prepared it into proper format. Further, cleaning of data performed to detect missing values and outliers in data file. At last but not least, we used binning techniques from large to small scale of data for better preparation and identification of rule after evaluation of ARM model. Data after preprocessing is enlisted in figure no. 2 below.

C. ASSOCIATION RULE MINING (ARM)

ARM algorithm is used to identify rules (frequent itemset) from large databases [11]. This work indicates implementation of association model evaluation for identification of features which may use for improvement of patients' health from disease schistosomiasis.

ARM algorithm is well-structured data mining technique [22]. Basic concept is to fulfil requirements of support (*supp.*) and confidence (*conf.*) for generation of a decision [11]–[22]. There are two sides of each rule indicated as left-hand side (LHS) and right-hand side (RHS). LHS are features from an attribute and RHS are generation of decision on the basis of LHS which are also consequences of another attribute [24]. Whenever LHS occurs, RHS will definitely be occurred or vice versa. Further, *supp.* and *conf.* are two rule measures that express rule significance and certainty [25]. Significance of a rule will be increased on the basis of *supp.* and *conf.* degree. There are many algorithms which have been used to generate association rules such as; Apriori, Tertius and Predictive Apriori (update version of Apriori) [11]–[26].

TABLE 3. Features' description.

No.	Attributes	No.	Attributes
1	Occupation	10	History of Ascites
2	Annual Income	11	Other Diseases
3	BMI (Body Mass Index)	12	The Extent of Ascites
4	Viability	13	Clinical Classification
5	Nourishment	14	Type of Treating Patients
6	Diagnostic Evidence1	15	Means of Treatment
7	Diagnostic Evidence2	16	Cost of Treatment
8	Prior Treatment	17	Class (Label)
9	History of Splenectomy		

Apriori technique is used in this research using *supp.* and *conf.* Further, below Table no 4 described the rules.

Apriori algorithm works as an iterative method which used to generate frequent item sets in given data set using X and Y variables and generate significant rules from it. For identification of association or rules between two item sets X and Y, necessity is to set the minimum support that is fraction of transactions which contains both X and Y called minsupp and minimum confidence for occurrence of both transactions as a combination (X and Y) and generation of rules on the basis of appearance of X variable, described in below equations.

$$\text{Supp.}(X \leq Y) = P(X \cap Y) \quad (\text{i})$$

$$\text{Conf.}(X \leq Y) = P(Y | X) = \frac{P(X \cap Y)}{P(X)} \quad (\text{ii})$$

IV. RESULTS AND DISCUSSION

Different algorithms were used to predict schistosomiasis disease in well and efficient manner and results of all algorithms were slightly different with own working criteria of every algorithm. In our research, primary aim was to investigate and analysis the disease deeply, assess associations between its factors, and validate it through association rules. Identification of factor which may become cause of individual's recovery and death from disease are examined by Apriori (ARM) method. We performed two approaches in this research. First approach was based on extraction of rules to indicate recovery of patient and second, to check factors which are causing the death of patients.

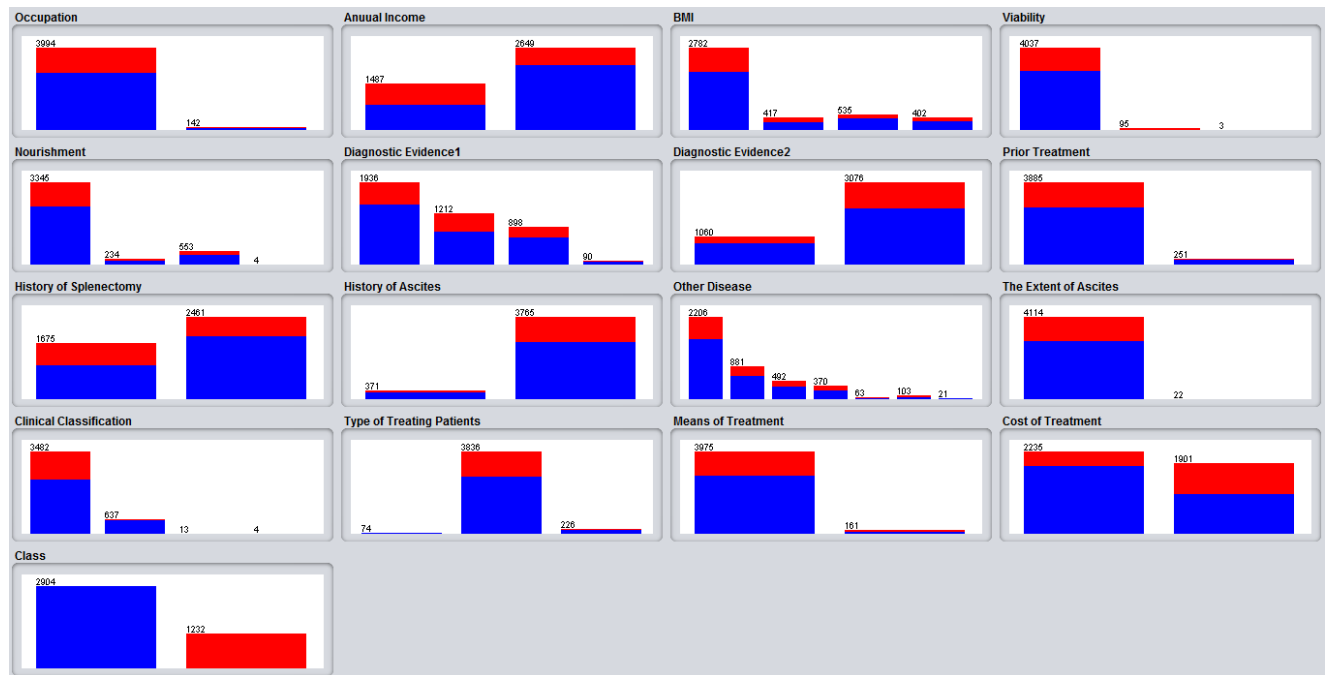


FIGURE 2. Visualization of dataset.

TABLE 4. Extraction of rules.

Algorithm	Rules
	Recovery Rules: If (The Extent of Ascites = Mild to Moderate) $\rightarrow$ Class = Recovery (conf., 1) If (Occupation = Farmer $\cap$ Prior Treatment = Yes $\cap$ The Extent of Ascites = Mild to Moderate) $\rightarrow$ Class = Recovery (conf., 1) If (Occupation = Farmer $\cap$ The Extent of Ascites = Mild to Moderate $\cap$ Clinical Classification = Megalosplenitis) $\rightarrow$ Class = Recovery (conf., 1) If (Prior Treatment = Yes $\cap$ The Extent of Ascites = Mild to Moderate $\cap$ Clinical Classification = Megalosplenitis) $\rightarrow$ Class = Recovery (conf., 1) Apriori If (BMI = Normal $\cap$ Viability = Normal $\cap$ Nourishment = Medium $\cap$ Prior Treatment = Yes $\cap$ Type of Treating Patients = Approved Retreated Patients) $\rightarrow$ Class = Recovery (conf., 1) If (BMI = Normal $\cap$ Nourishment = Medium $\cap$ Prior Treatment = Yes $\cap$ The Extent of Ascites = Mild to Moderate $\cap$ Clinical Classification = Megalosplenitis) $\rightarrow$ Class = Recovery (conf., 1) If (Annual Income = High $\cap$ Viability = Normal $\cap$ Prior Treatment = Yes $\cap$ The Extent of Ascites = Mild to Moderate $\cap$ Type of Treating Patients = Approved Retreated Patients) $\rightarrow$ Class = Recovery (conf., 1) Death Rules: If (Means of Treatment = Medical Treatment) $\rightarrow$ Class = Death (conf., 1) If (Diagnostic Evidence2 = Ascites Type $\cap$ Other Disease = No $\cap$ Cost of Treatment = High) $\rightarrow$ Class = Death (conf., 1)

In this research experiment, we used 4236 patient's data with disease schistosomiasis. First, we split all healthy samples in one class named as recovery and remaining (dead) samples into another class. After implementation of filtration and preprocessing techniques on data, we evaluate popular association rule mining algorithm (Apriori) using proposed

study framework. Generation of rules from given dataset are evaluated using minimum support (*minsupp*) and minimum confidence (*conf.*) which are key parameters of Apriori. Further, to generate rules from our given dataset, we set out *minsupp* (healthy rules) from 0.50 to 0.90 with 0.90 and above minimum *conf.* which gives us seven healthy rules from both

files. On the other hand, from second sample file, *minsupp* value consists of 0.90 and 0.25 for two and minimum *conf*. 0.90 or above.

From given results in below Table no. 4 it is observed that, some socioeconomic factors like occupation, annual income and prior treatment are varies a lot for improvement and sickness of a person's condition and important for treatment place but many others medical term features such as; ascites, extent of ascites, viability, nourishment, BMI etc. which may be handled at early stage of disease before in become danger for humans life, are known after factors identification. Detail of both approaches are described in table no. 4 below.

V. CONCLUSION

The capability to predict disease and identification of factors causing that disease at early stage, assumes as a significant part for patient's survival and appropriate treatment procedure. Data mining techniques are able now to diagnose disease using computerized learning approaches.

This research work has indicated association rule extraction analysis on schistosomiasis disease data set using rule mining method (Apriori). Further, association rule mining-based methods works on categorical data and our data was consists of different features and important risk factors in it.

Interestingly, healthy rules found from recovery rules, individuals have more chances to survive if they were affected and treated previously from schistosomiasis. Research also indicates that, if a person has BMI as normal, work ability is normal, previous treatment is also done etc., then chances of recovery are more. Further, results of Apriori method, hence, spotlighted useful knowledge.

Overall, this research work has illustrated the use of association rule mining (Apriori) method to direct interesting knowledge. Medically, professionals and practitioners are having disparity regarding highlighted factors. Our research work concentrates on the computerized applications, meanwhile, association rule mining method or rule-based classifiers, to spot key factors that are beyond the disease.

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